CASE 4

Mrs. Catherine Holt — 52F · SPMS | 2-year progressive decline · Rolling frame (all indoor mobility) · Bilateral adductor spasticity · Personal care and transfers impaired · Lhermitte's sign

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- • Significant bilateral lower limb weakness (rolling frame required for all indoor mobility)
- Bilateral lower limb spasticity — adductors predominant (UMN pattern, progressive)
- Significant bilateral lower limb weakness (rolling frame required for all indoor mobility)

■■ RED FLAGS

- 3. Tell me about Lhermitte's sign — is it longstanding or has it changed recently, what position triggers it, and how long does the sensation last?
- 4. Walk me through a typical transfer — for example, bed to chair and chair to toilet. At exactly which point does it break down, and what does it look like when it goes wrong?
- 5. Are you having any pain — particularly in the legs, hips, or lower back — and have you or your carer noticed any skin changes, redness, or pressure areas?
- 6. Who helps you with your care at home, and how are they managing with the transfers? Are there any concerns about their safety or yours during care?
- ■ Sudden or significantly worsening Lhermitte's sign
- Significance:
- Avoid all cervical flexion-loaded activities immediately. Urgent neurologist referral. MRI cervical spine may be required. Suspend neck-involving physiotherapy until reviewed.
- ■ Rapidly escalating spasticity over days or weeks

■ PATHOPHYSIOLOGY (KEY POINTS)

- ▼ Neurological Mechanism — SPMS Secondary Progressive MS represents a transition from the predominantly inflammatory, relapse-driven pathology of RRMS to a phase dominated by diffuse neurodegeneration and irreversible axonal loss .
- While focal inflammatory plaques still occur, the CNS repair mechanisms that enabled partial remission — remyelination by oligodendrocyte precursors, cortical plasticity, and functional reorganisation — become progressively overwhelmed.
- Chronically demyelinated axons, deprived of the trophic support normally provided by myelin, undergo Wallerian degeneration at an accelerating rate.
- Grey matter atrophy, reduction in total brain volume, and spinal cord atrophy all accumulate, explaining why symptoms become fixed and progressive rather than episodic.
- The bilateral adductor spasticity reflects upper motor neuron (UMN) pathway disruption in the bilateral corticospinal tracts at thoracic or lumbar spinal cord level.
- Loss of supraspinal inhibitory input to spinal interneurons leads to hyperexcitability of alpha-motor neurons supplying the hip adductors and hip flexors.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations

Tone: Observe bilateral lower limb tone at rest — scissoring posture of legs (sustained adductor pull), clonus at ankle on rapid dorsiflexion. Perform Modified Ashworth assessment of adductors in supine. Avoid sustained neck flexion throughout assessment (Lhermitte's precaution). Coordination: Heel-shin test from supine or seated — look for cerebellar ataxia component overlying the spasticity. Finger-nose if upper limb involvement present. Posture (seated):

Modified Ashworth Scale (MAS) — Bilateral Hip Adductors

Spasticity What it measures: Resistance to passive movement — grades spasticity on a 0–4 scale (0 = no increase; 1 = slight catch; 1+ = catch with minimal resistance; 2 = marked increase but passive movement possible; 3 = considerable increase, passive movement difficult; 4 = rigid). The standard clinical tool for quantifying and tracking spasticity. How to perform: Position Catherine supine on the plinth. If the transfer is too difficult, assess in her bed — document this adaptation.

Passive Range of Motion — Hip Abduction (Goniometry)

Contracture Screen What it measures: Available hip abduction range bilaterally, distinguishing flexible (dynamic) spasticity from fixed musculotendinous contracture. This distinction determines whether stretching is likely to be effective or whether surgical/orthopaedic input is needed. How to perform: Catherine supine; pelvis stabilised. Use a standard goniometer with the axis at the anterior superior iliac spine, stationary arm along the midline, and moving arm along the femur.

■ TREATMENT PLAN

- ■ Signs of deep vein thrombosis in lower limb

• Reduced mobility in SPMS significantly elevates DVT risk. Unilateral leg swelling, warmth, redness, and calf tenderness warrant urgent investigation. A DVT is also a major spasticity precipitant.
• Wells Score assessment; urgent GP/ED referral for Doppler ultrasound. Withhold passive stretching of the affected limb until excluded.
• Likely pain secondary to spasticity (groin, hip, adductor origin)
• Likely bladder / bowel dysfunction (spasticity precipitant + independent impairment)
• Functional Activity Limitations
• All transfers (bed, chair, toilet) compromised by spasticity and weakness
• Personal care limited — adductor spasticity closes legs, impeding perineal access
• All indoor mobility requires rolling frame; outdoor mobility likely absent or minimal
• Dependence on carer for most ADLs
• Participation Restrictions
• Loss of independence — significant impact on identity and dignity